

Vehicle Fuel Efficiency Prediction

Midterm project · Machine Learning Algorithms

Problem and goal

Question:

- Can we predict a new car's fuel consumption from its basic characteristics?

Target:

- Combined (L/100 km)

Task:

- Regression

Metrics:

- MAE (main), RMSE, R^2

Success criterion:

- Best model's MAE at least 2x lower than the baseline, $R^2 \geq 0.8$



Dataset

	Model	year	Make	Model	Vehicle class	\
0		2014	Acura	ILX	Compact	
1		2014	Acura	ILX	Compact	
2		2014	Acura	ILX Hybrid	Compact	
3		2014	Acura	MDX 4WD	Sport utility vehicle: Small	
4		2014	Acura	RDX AWD	Sport utility vehicle: Small	
	Engine size (L)	Cylinders	Transmission	Fuel type	City (L/100 km)	\
0	2.0	4	AS5	Z	9.9	
1	2.4	4	M6	Z	11.2	
2	1.5	4	AV7	Z	6.0	
3	3.5	6	AS6	Z	12.7	
4	3.5	6	AS6	Z	12.1	
	Highway (L/100 km)	Combined (L/100 km)	Combined (mpg)	\		
0	6.7	8.5	33			
1	7.7	9.6	29			
2	5.8	5.9	48			
3	9.1	11.1	25			
4	8.7	10.6	27			
	CO2 emissions (g/km)	CO2 rating	Smog rating			
0	196	NaN	NaN			
1	221	NaN	NaN			
2	136	NaN	NaN			
3	255	NaN	NaN			
4	244	NaN	NaN			

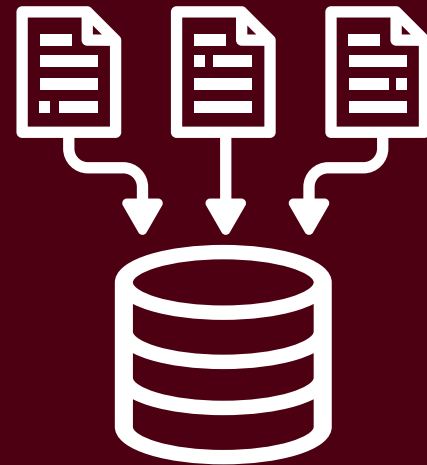
Source: Natural Resources Canada, Fuel Consumption Ratings

License: Open Government Licence - Canada

Size: 11829 cars, 15 columns, years 2014-2025

Features: year, make, model, vehicle class, engine size, cylinders, transmission, fuel type

Limitation: no vehicle weight or power



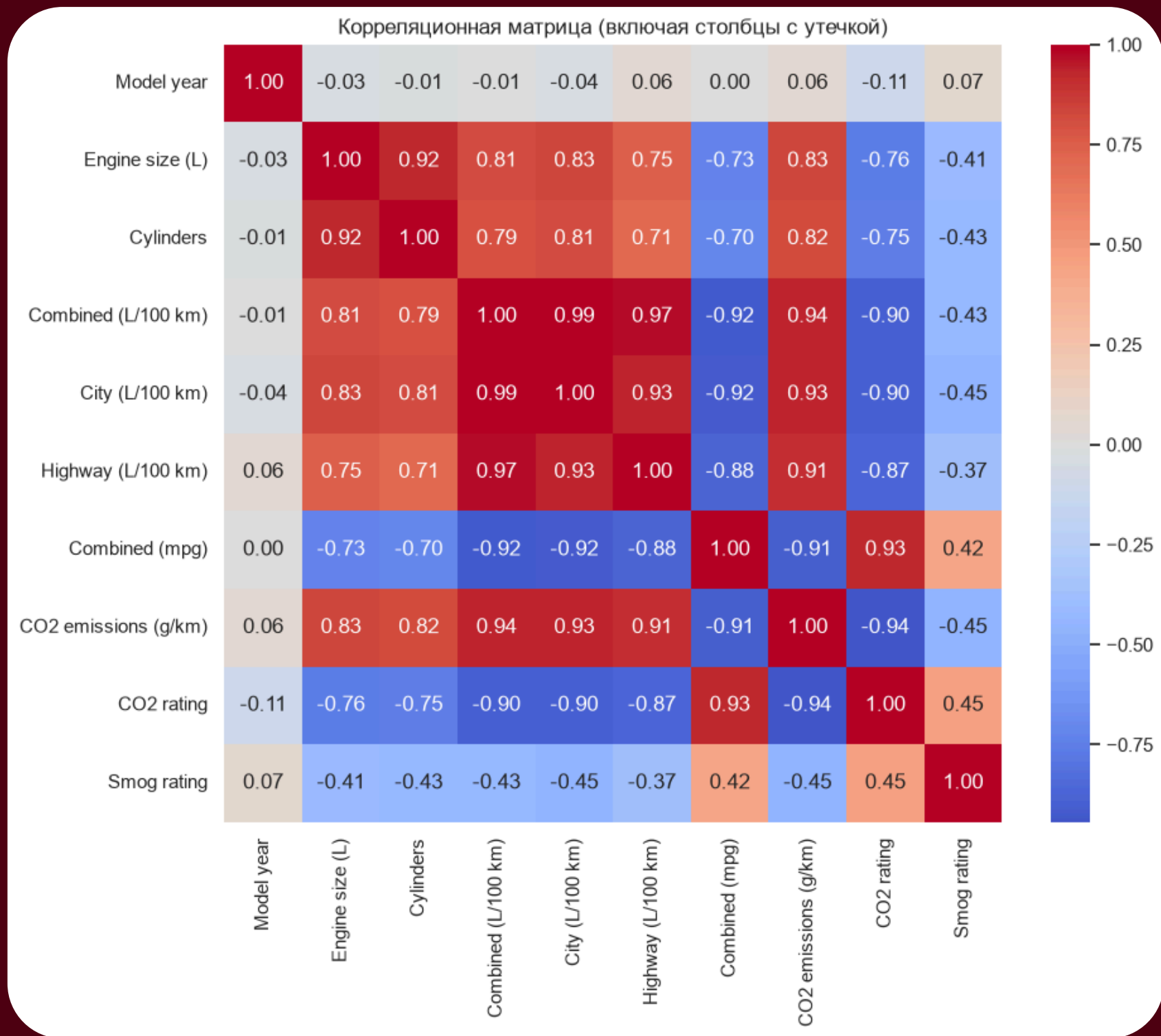
Data cleaning



- **No duplicates**
- **Missing values** only in CO2 rating and Smog rating: they depend on the year (not published for early years), so we do not fill them
- **Removed 1 row** with Fuel type = N (11829 → 11828 rows): one row cannot teach the model
- **Trimmed spaces and lowercased Model names** (2294 → 2279 unique names)
- **Outliers kept:** powerful SUVs and hybrids are real cars

	rows	co2_rating_missing	smog_rating_missing
Model year			
2014	1068	1068	1068
2015	1128	1128	1128
2016	1106	0	1106
2017	1058	0	0
2018	1083	0	0
2019	1056	0	0
2020	975	0	0
2021	970	0	0
2022	976	0	0
2023	919	0	0
2024	789	0	0
2025	701	0	0

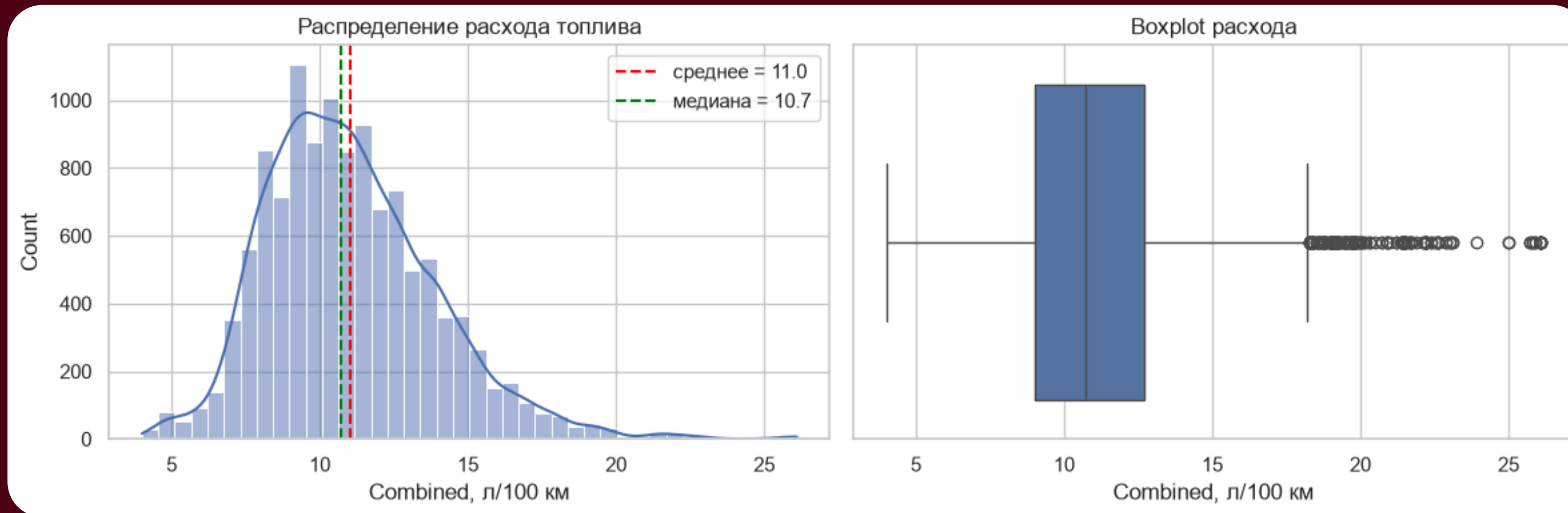
Data leakage



- 6 columns directly express the target and are excluded: City, Highway, Combined (mpg), CO2 emissions, CO2 rating, Smog rating
- Correlation with target: 0.90-0.99 for five of them
- ***Without this, the model would get the answer as a hint***

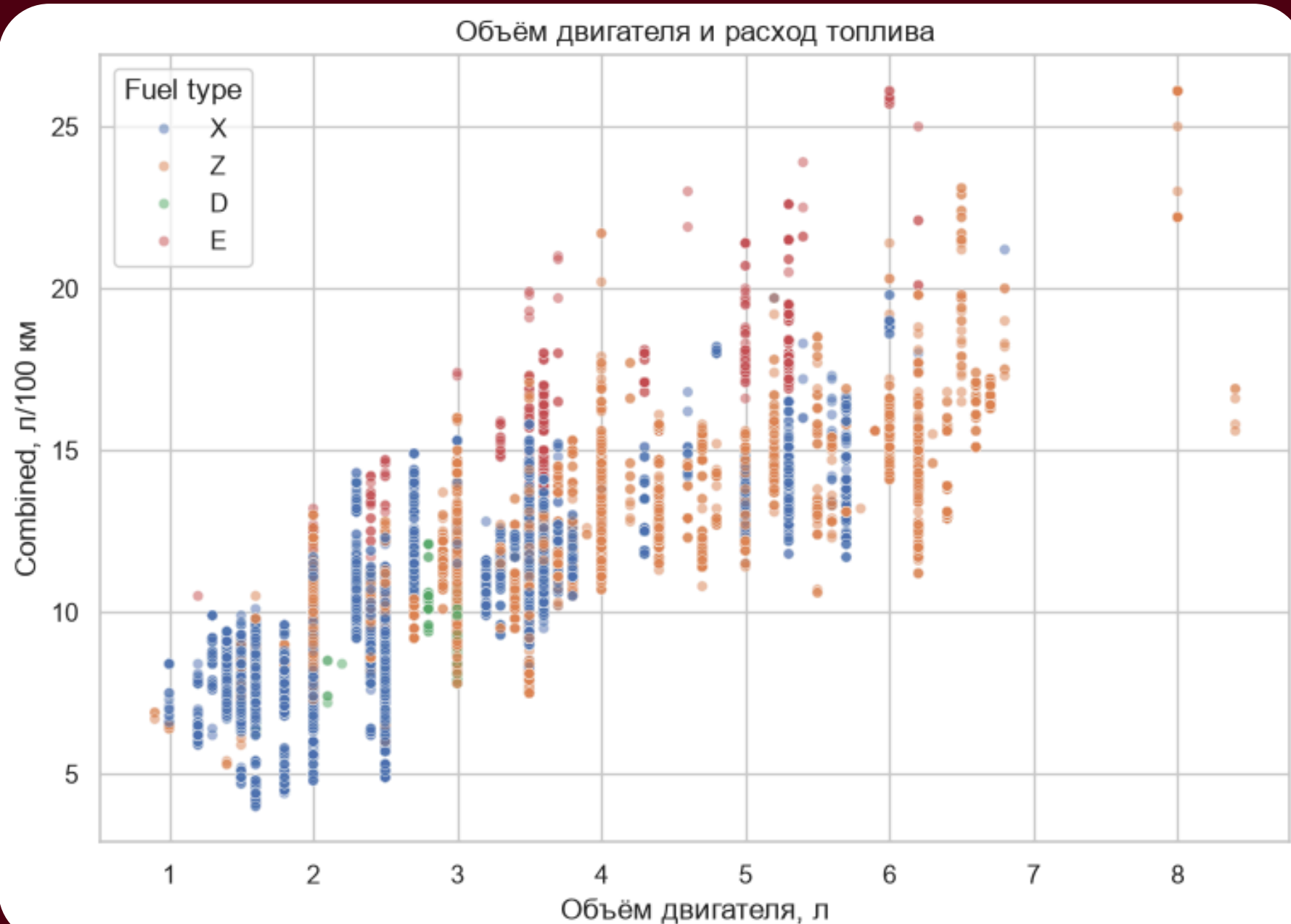


EDA: key findings



- Target is right-skewed (skew 0.81): mean 11.03, median 10.7, tail up to 26.1
- Bigger engine means higher consumption (correlation 0.81)
- Fuel type shifts consumption: E85 17.0, premium 11.6, regular 10.1, diesel 9.3
- Model year is almost unrelated to the target (-0.01)

EDA: key findings



```
plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=df, x="Engine size (L)", y=TARGET,
    hue="Fuel type", hue_order=["X", "Z", "D", "E"],
    alpha=0.5, s=20
)
plt.title("Объём двигателя и расход топлива")
plt.xlabel("Объём двигателя, л")
plt.ylabel("Combined, л/100 км")
plt.show()

print("Корреляция Engine size и target:",
      round(df["Engine size (L)"].corr(df[TARGET]), 3))
```

Feature engineering and split

```
from sklearn.model_selection import train_test_split

# Обычный случайный split, для сравнения
g_a, g_b = train_test_split(groups, test_size=0.15, random_state=42)

# Какая доля строк test имеет такую же модель в train
print("Обычный split:", round(g_b.isin(g_a).mean() * 100, 1), "%")
print("Group split:", round(g_test.isin(g_train).mean() * 100, 1), "%")
```

Обычный split: 94.7 %
Group split: 0.0 %

- Group split by Make + Model: train 70% (8268 rows) / validation 15% (1809) / test 15% (1751)
- Why: the same model appears in different years and would leak into test
- Random split: 94.7% of test rows have a twin in train; group split: 0%

- Features: Transmission split into trans_type and gears; new feature disp_per_cyl; median imputation; one-hot encoding
- No feature scaling (project requirement)

```
from sklearn.model_selection import GroupShuffleSplit, GroupKFold

RANDOM_STATE = 42
groups = df["Make"] + " | " + df["Model"].str.strip()

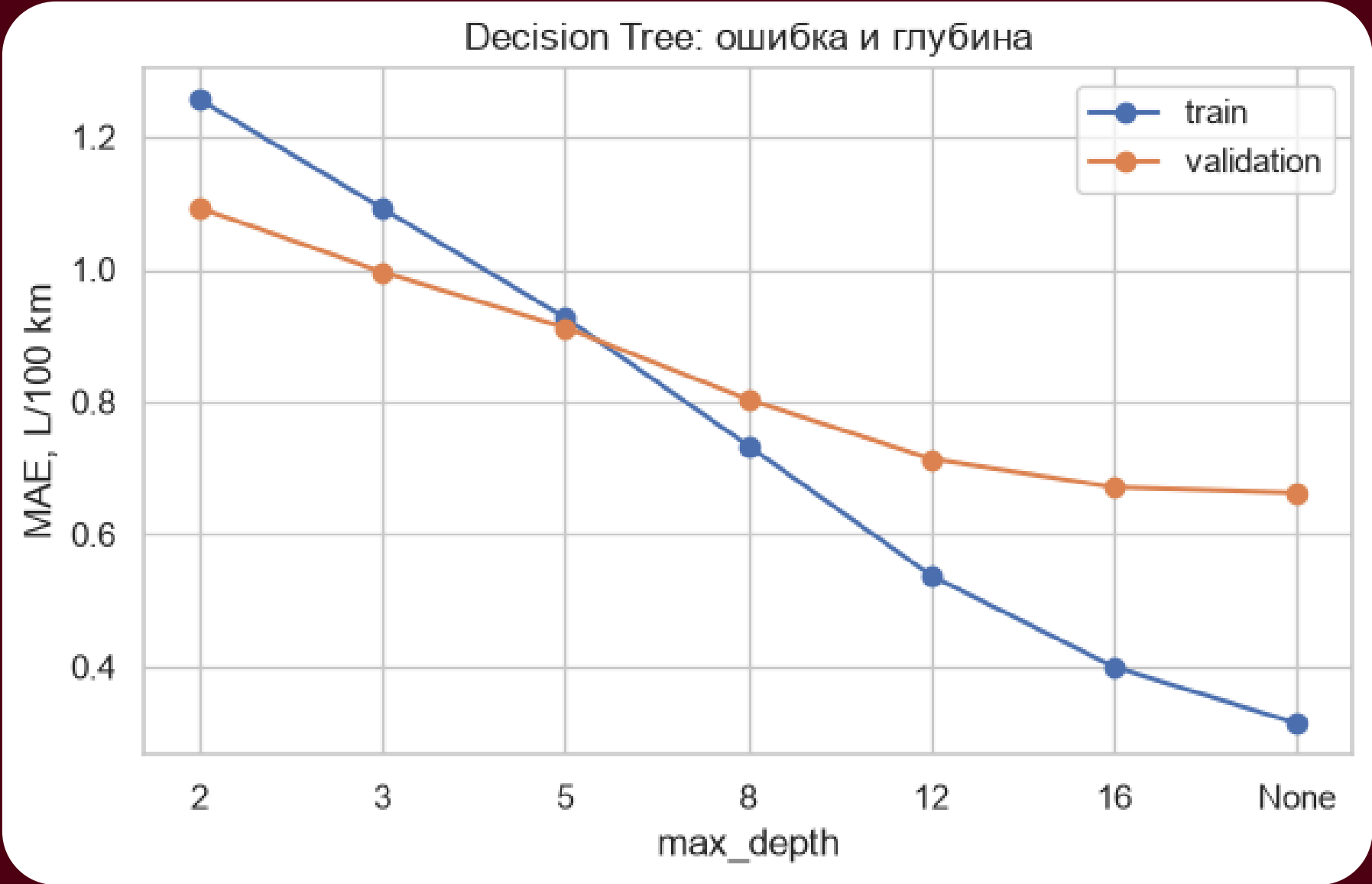
# Шаг 1: отделяем test (15%)
split1 = GroupShuffleSplit(n_splits=1, test_size=0.15, random_state=RANDOM_STATE)
temp_idx, test_idx = next(split1.split(X, y, groups))
X_temp, y_temp, g_temp = X.iloc[temp_idx], y.iloc[temp_idx], groups.iloc[temp_idx]
X_test, y_test, g_test = X.iloc[test_idx], y.iloc[test_idx], groups.iloc[test_idx]

# Шаг 2: из остатка отделяем validation (15% от всех данных)
split2 = GroupShuffleSplit(n_splits=1, test_size=0.15 / 0.85, random_state=RANDOM_STATE)
train_idx, val_idx = next(split2.split(X_temp, y_temp, g_temp))
X_train, y_train, g_train = X_temp.iloc[train_idx], y_temp.iloc[train_idx], g_temp.iloc[train_idx]
X_val, y_val, g_val = X_temp.iloc[val_idx], y_temp.iloc[val_idx], g_temp.iloc[val_idx]

print("train:", len(X_train), "| val:", len(X_val), "| test:", len(X_test))
```

Models and baseline

- Baseline: predicts the train mean (validation MAE 2.123)
- Models: Linear Regression (0.802), Decision Tree (0.662), KNN (0.746), all MAE on validation
- Hyperparameters chosen on validation: tree max_depth=None, KNN k=5
- Tree overfits: train MAE 0.314 vs validation 0.662
- Cross-validation (GroupKFold, 5 folds), MAE: Tree 0.736 ± 0.040 , KNN 0.816 ± 0.019 , Linear 0.880 ± 0.049 . The order of models is stable
- No feature scaling, so KNN is weaker (MAE 0.667 with scaling in a trial run)

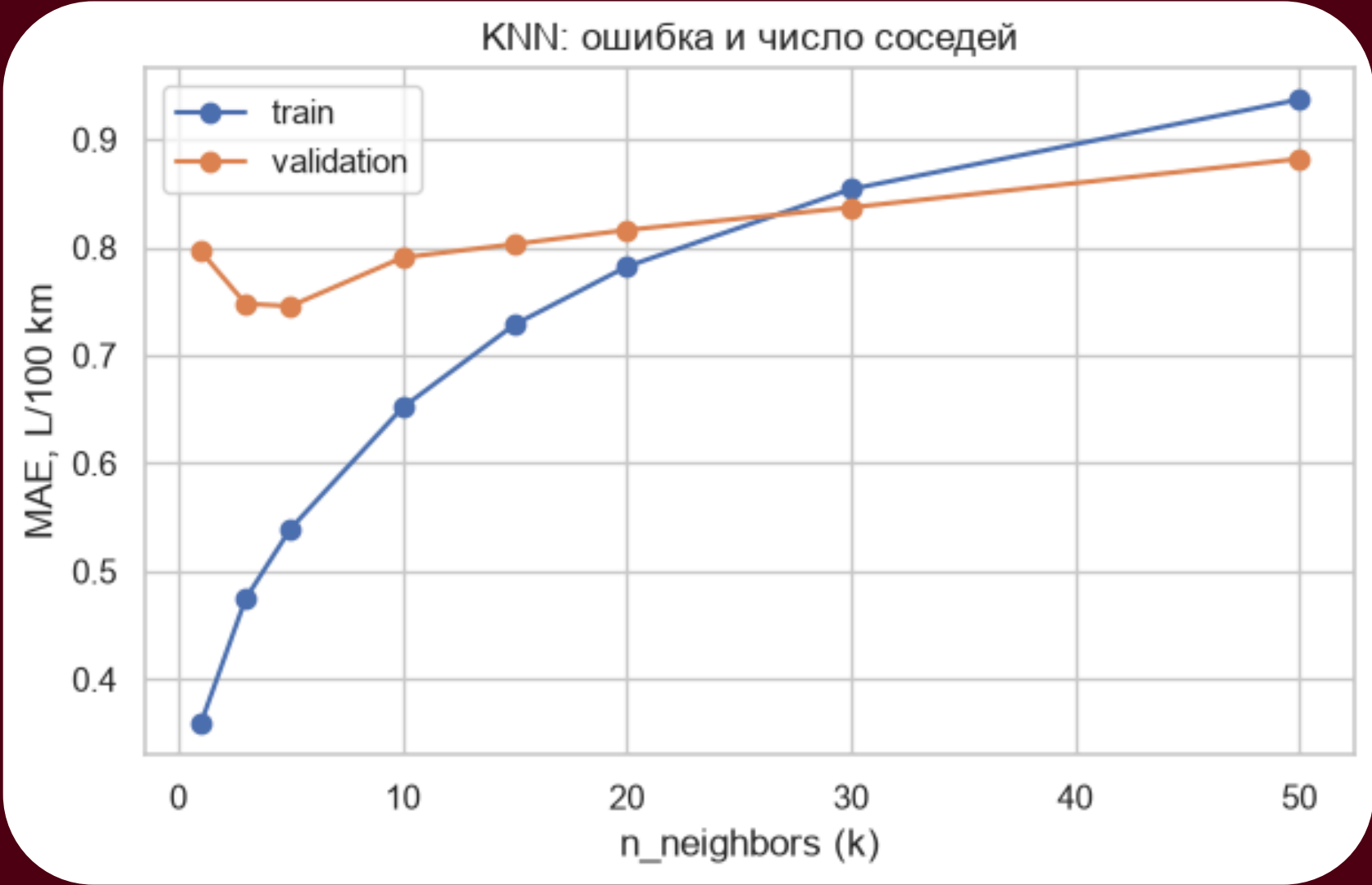


	max_depth	MAE_train	MAE_val
0	2	1.257	1.093
1	3	1.092	0.996
2	5	0.927	0.912
3	8	0.734	0.803
4	12	0.538	0.714
5	16	0.400	0.673
6	None	0.314	0.662

Best depth: None			
	MAE	RMSE	R2
Mean baseline	2.123	2.663	-0.000
Linear Regression	0.802	1.074	0.837
Decision Tree	0.662	0.963	0.869

Models and baseline

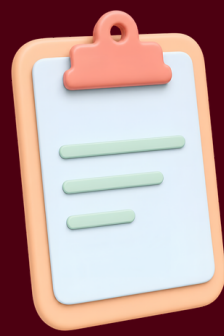
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	k	MAE_train	MAE_val
0	1	0.359	0.796
1	3	0.474	0.748
2	5	0.539	0.746
3	10	0.652	0.791
4	15	0.729	0.803
5	20	0.782	0.816
6	30	0.854	0.837
7	50	0.937	0.882

Best k: 5			
	MAE	RMSE	R2
Mean baseline	2.123	2.663	-0.000
Linear Regression	0.802	1.074	0.837
Decision Tree	0.662	0.963	0.869
KNN	0.746	1.037	0.848

Results



Best: Decision Tree, MAE 3.52x lower than the baseline

Success criterion: met (MAE ratio ≥ 2 , $R^2 \geq 0.8$)

Same model order on validation, CV and test

Test MAE (0.628) is lower than CV (0.736): the final models are trained on train + validation, and the test groups may be easier

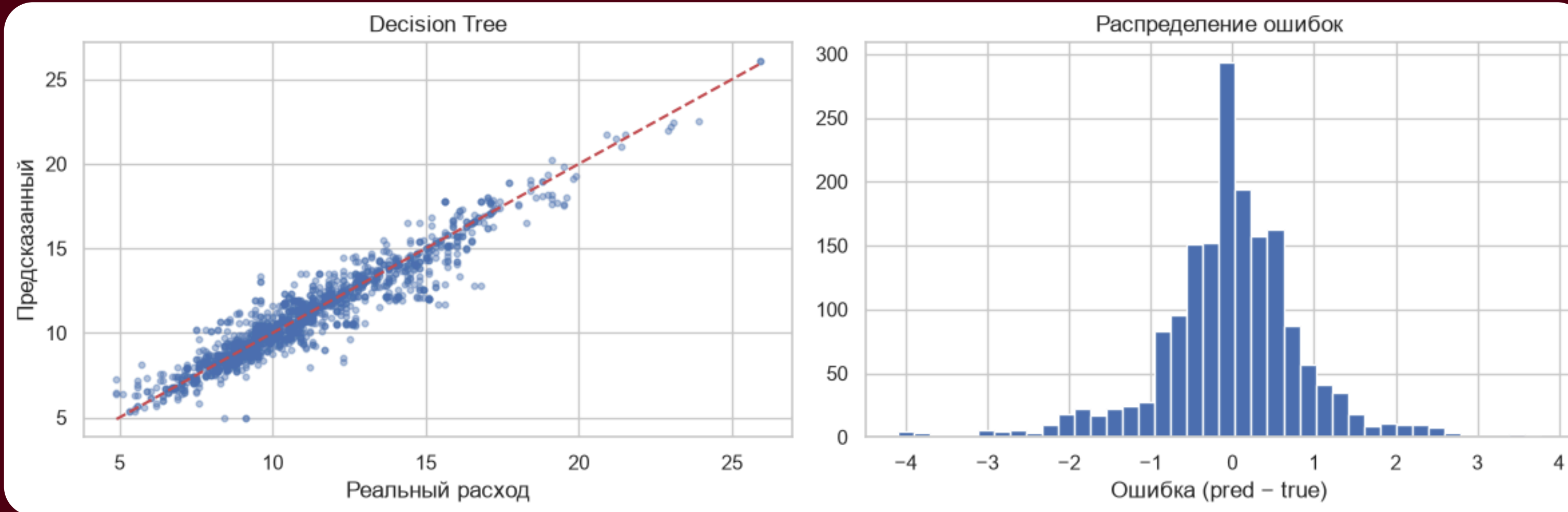
	MAE	RMSE	R2
Mean baseline	2.211	2.835	-0.000
Linear Regression	0.780	1.028	0.868
Decision Tree	0.628	0.920	0.895
KNN	0.715	0.973	0.882

Best model: Decision Tree

MAE baseline / best MAE = 3.52 (need ≥ 2)

Best R^2 = 0.895 (need ≥ 0.8)

Error analysis, open problems, next steps



Errors: hybrids overestimated by +0.96 L/100 km (MAE 1.03 vs 0.60 for the rest, 45 cars in test)

“Supercharged” cars (Camaro ZL1, Lancer Evolution) underestimated by 3.7-4.0 L/100 km

Open problems: tree overfitted; no hybrid feature; no power or weight in the data; rare groups (5 and 16 cylinders, Van: Cargo)

Plan for the final: GridSearchCV with GroupKFold (max_depth, min_samples_leaf), is_hybrid feature, time-based check (train 2014-2023, test 2024-2025), KNN without Model year (0.715 vs 0.746 on validation)



The End